Optimisation of a Grid-Connected Solar Photovoltaic–Battery Microgrid for Economic Energy Management

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Abstract: The increasing penetration of renewable energy sources has accelerated the deployment of grid-connected microgrids as a sustainable solution for future power systems. However, conventional energy management strategies for photovoltaic (PV)-battery microgrids primarily focus on minimizing operating costs or maximizing self-consumption, often overlooking the potential contribution of distributed energy resources to utility grid support. This research proposes a grid-support-oriented multi-objective optimization framework for a grid-connected solar PV-battery-load microgrid. The optimization problem aims to simultaneously minimize electricity imported from the utility grid, maximize renewable energy exported to the grid, maximize the utilization of locally generated renewable energy, maximize the overall profit of the microgrid, and minimize system operating costs while satisfying power balance and battery operational constraints. The proposed optimization model will first be formulated and validated using representative load demand and solar generation profiles. Subsequently, the framework will be extended to a campus-scale case study using real electricity demand and photovoltaic generation data from the University of Tsukuba. Future work will investigate the integration of Vehicle-to-Grid (V2G) technology to further enhance energy flexibility and grid support services. The proposed framework is expected to improve both the economic performance of renewable microgrids and the operational efficiency of future smart grids.

Keywords: microgrid,optimisation problem, minimising grid import, maximising microgrid export, optimal economy problem, battery control, university campus microgrid simulation, self-sufficiency, self-consumption, PV, peak shifting, demand response

1 Introduction

There is increasing need for integration of renewable en ergy source (RES) and energy storage (RESS) into electrical power systems due to environmental and sustainability concerns. One of the ways this can be achieved is by adding RES and RESSS into a microgrid. A micro-grid is defined as a combination of a source of electricity and a load, both of which have been localized. A MG is connected to main grid at the point of common coupling (PCC). It can function in two modes—either autonomously or in synchronization with the main grid. The operation of a MG is very similar to that of the main grid with the exception that it is meant for a smaller geographical area. It operates as a single controllable entity. [1]

Microgrids are being used to improve reliability and resilience of electrical grids, to manage the addition of distributed clean energy sources such as solar photovoltaic (PV) to reduce fossil fuel emissions, and to provide electricity in areas not served by centralised electrical structure. Microgrids, which are composed of distributed generation systems, energy storage, and loads, are being developed in various areas for both low and medium voltage ranges. [3]

The increasing penetration of renewable energy sources has become an essential strategy for reducing greenhouse gas emissions and achieving global carbon neutrality targets. Among various renewable energy technologies, solar photovoltaic (PV) systems have experienced significant growth because of their decreasing installation costs and environmental benefits. According to the International Energy Agency, solar PV is expected to become one of the largest contributors to future electricity generation, requiring more advanced energy management strategies to address its intermittent nature.

Battery Energy Storage Systems (BESS) have become indispensable components of renewable-energy-based microgrids. Batteries mitigate the variability of solar generation by storing surplus energy during periods of high irradiation and supplying electricity when renewable generation is insufficient. Consequently, numerous studies have investigated optimal battery scheduling methods to reduce electricity costs, improve self-consumption, or extend battery lifetime.

The transition toward low-carbon energy systems has significantly increased the deployment of renewable energy resources, particularly solar photovoltaic (PV) systems, in residential, commercial, and institutional power networks. Although renewable energy contributes to reducing greenhouse gas emissions, its intermittent nature introduces operational challenges, including fluctuations in power generation, increased dependence on the utility grid during periods of low renewable output, and renewable energy curtailment during periods of surplus generation. Consequently, Battery Energy Storage Systems (BESS) have become essential components of modern microgrids by providing operational flexibility through energy storage and dispatch.

Conventional Energy Management Systems (EMS) for grid-connected PV-battery microgrids primarily optimize battery charging and discharging schedules to minimize electricity costs, maximize self-consumption of locally generated renewable energy, or extend battery lifetime. While these strategies improve the economic performance of the microgrid owner, they generally treat the utility grid as a passive energy supplier and overlook the role that distributed renewable resources can play in supporting future smart grids.

As renewable penetration continues to increase, distributed batteries are expected to evolve beyond local energy storage

devices into active grid-support resources. Intelligent battery scheduling can reduce dependence on imported electricity, increase renewable energy supplied to the utility grid during periods of surplus generation, and improve renewable energy utilization across the distribution network. However, such strategies must also remain economically attractive to the microgrid owner by considering operating costs and potential revenue obtained through electricity export.

Motivated by these challenges, this research proposes a grid-support-oriented multi-objective optimization framework for a grid-connected solar PV-battery-load microgrid. The proposed optimization simultaneously pursues five objectives:

1. Minimize electricity imported from the utility grid and maximize the utilization of locally generated renewable energy
2. Maximize renewable electricity exported to the utility grid.
3. Maximize the overall profit of the microgrid.
4. Minimize the total operating cost of the microgrid.

The first and second objective functions are motivated by the economic interaction between the microgrid and the utility grid. Electricity imported from the utility grid represents an operating expense for the microgrid owner, while the corresponding payment constitutes revenue for the utility. Therefore, minimizing electricity purchases reduces the microgrid's operating cost. Conversely, when surplus renewable energy or stored battery energy is available after satisfying local demand, exporting this excess electricity to the utility grid generates additional revenue for the microgrid. Consequently, the proposed optimization simultaneously minimizes electricity imported from the grid and maximizes profitable electricity exports to improve the overall economic performance of the microgrid.

These objectives are optimized while satisfying power balance, battery operational limits, inverter capacity, and grid exchange constraints. Unlike conventional energy management strategies that prioritize self-consumption, the proposed framework seeks to balance the economic interests of the microgrid operator with the operational needs of the utility grid through cooperative battery scheduling.

Peak shifting is included to improve the economic performance of the microgrid while enhancing renewable energy utilization. Instead of purchasing electricity during periods when locally generated renewable energy is available, excess photovoltaic energy can be stored in the battery or exported to the utility grid. During periods of high electricity demand or reduced renewable generation, the stored energy can be discharged to supply the local load, thereby reducing electricity imports and operating costs. Furthermore, shifting energy exchange toward daytime periods allows surplus solar generation to be exported when renewable energy is readily available, increasing microgrid revenue and contributing to grid support.

The research will be conducted in two phases. The first phase focuses on developing the mathematical optimization formulation and validating the proposed Energy Management System using representative photovoltaic generation and load demand profiles. The second phase extends the optimization framework to a campus-scale case study using real electricity demand and photovoltaic generation data from the University of Tsukuba Electricity Monitoring System (TEMS)to evaluate its feasibility in a practical renewable energy microgrid. Finally, the proposed framework is designed to accommodate future integration of Vehicle-to-Grid (V2G) technology, allowing electric vehicles to participate as distributed energy storage resources that further enhance renewable energy utilization, microgrid profitability, and grid support capabilities.

By integrating technical, economic, and grid-support objectives within a unified optimization framework, this research aims to contribute to the development of next-generation smart microgrids that benefit both microgrid operators and the wider electricity network.

Currently, we are formulating the optimisation model. The objective functions have been decided. However, the mathematical formulas of the objective functions, decision variables, and constraints have not yet been finalised and formulated.

2 Literature Review

The increasing deployment of renewable energy technologies has accelerated the development of microgrids as an effective solution for improving energy sustainability, energy security, and grid resilience. A microgrid is a localized energy system comprising distributed energy resources (DERs), energy storage systems, and electrical loads that can operate either in grid-connected or islanded modes [3], [4]. Owing to their operational flexibility, microgrids have been widely investigated for residential, commercial, industrial, and university campus applications [4].

Among various renewable energy technologies, solar photovoltaic (PV) systems have become one of the most rapidly expanding renewable generation sources due to their declining installation costs and environmental benefits. However, PV generation is inherently intermittent because it depends on solar irradiance and weather conditions, resulting in fluctuations in power generation that complicate energy management and grid operation [5]. Consequently, intelligent Energy Management Systems (EMS) are required to coordinate renewable generation, energy storage, and utility grid interaction while maintaining reliable system operation.

University campuses have recently emerged as attractive platforms for renewable-energy-based microgrids because of their relatively predictable load characteristics, large rooftop areas suitable for PV installations, and institutional commitments toward carbon neutrality [6]. These characteristics make campus microgrids valuable real-world testbeds for evaluating advanced optimization-based energy management strategies before deployment at the city or distribution-network level. The application of optimization-based energy management has also been extensively investigated using real campus energy systems. Michael Stadler et al. developed an optimization framework for a university campus microgrid incorporating distributed energy resources, combined heat and power (CHP), photovoltaic systems, battery energy storage, and thermal storage. The optimization was formulated us-

ing mixed-integer linear programming (MILP) to determine the optimal operation and scheduling of distributed energy resources while minimizing the overall operating cost and improving system efficiency under actual campus electricity and thermal demand profiles. Their results demonstrated that optimization-based scheduling can significantly reduce energy costs, improve renewable energy integration, and enhance the operational flexibility of campus-scale energy systems. Furthermore, the study showed that real campus load data provide a realistic platform for evaluating advanced energy management strategies before implementation in larger smart distribution networks.

Although the work of Stadler et al. successfully demonstrated the effectiveness of optimization using actual campus operational data, the primary objective was economic dispatch through operating cost minimization and efficient utilization of distributed energy resources. Electricity exported to the utility grid was not explicitly formulated as a principal optimization objective, and battery scheduling was not designed from a grid-support perspective. Consequently, opportunities to simultaneously improve microgrid profitability, increase renewable electricity exported to the utility grid, and maximize renewable energy utilization remain relatively unexplored.

Battery Energy Storage Systems (BESS) play a crucial role in mitigating the variability of renewable energy generation by storing excess electricity during periods of high solar output and supplying electricity during periods of insufficient renewable generation [5]. The integration of batteries significantly improves the operational flexibility, reliability, and economic performance of grid-connected microgrids.

Various optimization methods have been proposed for battery scheduling, including Mixed-Integer Linear Programming (MILP), Dynamic Programming (DP), Model Predictive Control (MPC), Particle Swarm Optimization (PSO), Genetic Algorithms (GA), and Differential Evolution (DE) [7], [9]. These optimization techniques generally aim to minimize electricity costs, reduce peak demand, maximize self-consumption of renewable energy, and prolong battery lifetime.

For example, Kim et al. proposed a hybrid solar–battery–fuel cell energy hub optimized using the cDEAS algorithm to minimize operating cost while satisfying system constraints [6]. The objective was to minimize the total operating cost by optimally distributing energy among the available power sources while satisfying power balance and storage constraints. The model was validated using 100 days of actual residential load and solar generation data from Korea, demonstrating an average operating cost reduction of 19.35% compared with conventional power-sharing methods. The authors concluded that optimization significantly improves the economic operation of hybrid renewable energy systems and suggested incorporating additional realistic cost factors and battery scheduling strategies in future work.Their study demonstrated that coordinated operation of multiple energy storage devices can improve overall system efficiency and reduce electricity costs. However, the optimization primarily focuses on economic dispatch from the perspective of the microgrid owner.

Similarly, Olivares et al. reviewed state-of-the-art microgrid control strategies and concluded that most existing EMS formulations prioritize minimizing operational costs while maintaining supply reliability [5]. Although these approaches successfully improve local microgrid operation, electricity exported to the utility grid is generally treated as a secondary outcome rather than a primary optimization objective.

As renewable penetration continues to increase, distributed batteries are expected to evolve from passive storage devices into active resources capable of supporting utility grid operation. Intelligent battery scheduling can reduce electricity imported from the utility grid, mitigate renewable energy curtailment, and export surplus renewable energy to improve renewable penetration at the distribution level [2].

Recent studies have investigated coordinated energy management involving photovoltaic systems, battery storage, hydrogen storage, and electric vehicles. Yu et al. proposed an optimization framework incorporating Vehicle-to-Grid (V2G) technology to coordinate renewable generation and EV charging within a microgrid [7]. Teir model coordinates renewable generation, electric vehicle charging/discharging, hydrogen storage, and demand response using a Lyre Bird Optimization Algorithm (LBOA). The optimization minimizes operating, environmental, and reliability costs while maximizing system revenue and maintaining high reliability under renewable energy uncertainty. Simulation results demonstrated significant reductions in total operating cost together with increased revenue from V2G participation and electricity sales to the grid. Their results demonstrated that coordinated optimization significantly improves renewable energy utilization and system flexibility. his study demonstrates that integrating V2G into renewable energy microgrids can substantially improve both economic performance and system flexibility. However, the proposed framework relies on a relatively complex microgrid architecture incorporating wind generation, hydrogen storage, fuel cells, demand response, and electric vehicle parking systems. The present research intentionally focuses on a simpler solar PV–battery–load microgrid, allowing the battery scheduling strategy itself to be investigated in greater depth. Vehicle-to-Grid functionality is reserved as a future extension after validating the proposed optimization framework using battery storage alone.

2.1 Microgrid Energy Markets and Economic Energy Management

Likewise, Mengelkamp et al. highlighted that future microgrids should not only optimize local energy consumption but also actively participate in electricity markets through intelligent energy trading and coordinated management of distributed energy resources [5]. To demonstrate this concept, the authors investigated the Brooklyn Microgrid project, proposing a decentralized market framework in which distributed energy resources, including rooftop photovoltaic systems and battery storage, can participate in local energy trading rather than relying solely on the utility grid as the electricity supplier [10]. The study identified seven essen-

tial components required for an effective microgrid energy market: (i) market mechanism, (ii) pricing mechanism, (iii) energy management system, (iv) information system, (v) communication infrastructure, (vi) regulatory framework, and (vii) physical power infrastructure. Using the Brooklyn Microgrid as a case study, the authors demonstrated that local energy trading enables prosumers to sell surplus renewable electricity while allowing consumers to purchase locally generated energy through a transparent market mechanism. Consequently, this approach improves renewable energy utilization, enhances economic efficiency, promotes distributed renewable generation, and reduces dependence on centralized electricity production [10].

The work of Mengelkamp et al. represents an important advancement in the economic operation of renewable-energy-based microgrids by recognizing electricity as a tradable commodity rather than merely a means of satisfying local demand. Their findings suggest that future Energy Management Systems (EMS) should integrate technical performance, economic benefits, and grid interaction within a unified operational framework. However, the study primarily focuses on market design and peer-to-peer energy trading, with comparatively less emphasis on the optimal scheduling of Battery Energy Storage Systems (BESS) under multiple technical and economic objectives. In particular, battery charging and discharging decisions are not explicitly optimized to maximize renewable electricity exported to the utility grid while simultaneously minimizing electricity imports and operating costs. These limitations motivate the present research, which develops a grid-support-oriented multi-objective optimization framework that integrates intelligent battery scheduling with electricity trading to simultaneously minimize grid imports, maximize renewable electricity exports, maximize renewable energy utilization, maximize microgrid profit, and minimize operating costs.

Nevertheless, existing studies primarily prioritize maximizing self-consumption and minimizing electricity costs for the microgrid owner. Comparatively fewer studies explicitly formulate battery scheduling strategies that prioritize exporting surplus renewable energy while simultaneously considering both microgrid profitability and utility grid support.

Despite significant progress in renewable energy microgrid optimization, several research gaps remain.

First, existing battery scheduling strategies predominantly emphasize minimizing electricity costs and maximizing self-consumption from the perspective of the microgrid owner [5], [7]. Consequently, the broader contribution of distributed batteries toward supporting utility grid operation has received comparatively limited attention.

Second, while electricity export is frequently included within power balance equations, it is rarely formulated as a primary optimization objective. Most optimization frameworks prioritize local consumption before exporting excess renewable energy, potentially overlooking opportunities to improve renewable energy utilization and grid flexibility.

Third, although campus microgrids have received increasing research attention, relatively few studies have developed optimization frameworks intended for validation using real campus electricity demand and photovoltaic generation data. Furthermore, future integration of Vehicle-to-Grid (V2G) technology is often investigated separately rather than incorporated as a natural extension of battery-based microgrid optimization.

3 Optimisation of Energy Systems

From a review of optimisation of energy systems by Frangopoulos, optimization is the act of obtaining the best result under given circumstances and mathematical optimization is the process of finding the conditions that give the minimum or maximum of a function, called the "objective function." [11]

3.1 Levels of energy systems optimization

The optimization of an energy system can be considered at three levels

A. *Synthesis optimization.* The term "synthesis" implies the components that appear in a system and their interconnections. After the synthesis of a system has been composed, the flow diagram of the system can be drawn.
B. *Design optimization.* The word "design" here is used to imply the technical characteristics (specifications) of the components and the properties of the substances entering and exiting each component at the "design" point (nominal load) of the system. One may argue that design includes synthesis too. However, in order to distinguish the various levels of optimization, the word "design" will be used with the particular meaning given here.
C. *Operation optimization.* For a given system (i.e., one in which the synthesis and design are known) under specified conditions, the optimal operating point is requested, as it is defined by the operating properties of components and substances in the system (speed of revolution, power output, mass flow rates, pressures, temperatures, composition of fluids, etc.).

Of course, if complete optimization is the goal, each level cannot be considered in complete isolation from the others. Consequently, the complete optimization problem can be stated by the following question:

> What is the synthesis of the system, the design characteristics of the components, and the operating point (or operating strategy) that lead to an overall optimum?

3.2 The time in optimization of energy systems

[8] classifies the energy-system optimization according to whether and how time is represented. It progresses from a single steady-state operating point to full life-cycle optimization containing both steady-state and transient periods. The five types are:

1. Static optimization
2. Dynamic optimization
3. Intertemporal static optimization

4. Optimization of transient conditions or optimal control
5. Dynamic synthesis-design-operation optimization including transients

FOr simplicity and relevant to our research project, the first three types will be discussed.

3.3 Static Optimization

In static optimization, time is not explicitly considered. The energy system is assumed to operate under steady-state conditions at a single operating point, which is usually the nominal or design point.

The general static optimization problem is written as

$$\underset{\mathbf{x}}{\text{minimize}} \quad f(\mathbf{x}), \tag{1}$$

where the independent-variable vector is

$$\mathbf{x} = (x_1, x_2, \ldots, x_n), \tag{2}$$

subject to the equality constraints

$$h_i(\mathbf{x}) = 0, \qquad i = 1, 2, \ldots, m, \tag{3}$$

and the inequality constraints

$$g_j(\mathbf{x}) \leq 0, \qquad j = 1, 2, \ldots, p. \tag{4}$$

For energy-system optimization, the independent variables can be divided into three groups:

$$\mathbf{x} \equiv (\mathbf{v}, \mathbf{w}, \mathbf{z}), \tag{5}$$

where

- $\mathbf{v}$ denotes operation variables, such as component load factors, mass-flow rates, pressures, and temperatures;
- $\mathbf{w}$ denotes design variables, such as nominal component capacities and design-point stream conditions;
- $\mathbf{z}$ denotes synthesis variables defining whether components and interconnections are included in the system.

The complete synthesis–design–operation optimization problem is therefore

$$\underset{\mathbf{v},\mathbf{w},\mathbf{z}}{\text{minimize}} \quad f(\mathbf{v}, \mathbf{w}, \mathbf{z}). \tag{1a}$$

For a specified system synthesis, meaning that $\mathbf{z}$ is fixed, the problem is reduced to design and operation optimization:

$$\underset{\mathbf{v},\mathbf{w}}{\text{minimize}} \quad f_d(\mathbf{v}, \mathbf{w}). \tag{1b}$$

When both the synthesis and design are fixed, only the operation variables are optimized:

$$\underset{\mathbf{v}}{\text{minimize}} \quad f_{\text{op}}(\mathbf{v}). \tag{1c}$$

A maximization problem can also be expressed as a minimization problem:

$$\min_{\mathbf{x}} f(\mathbf{x}) = \max_{\mathbf{x}} \{-f(\mathbf{x})\}. \tag{6}$$

The principal solution methods for static optimization are:

1. *Search methods*, which evaluate the objective function at different combinations of the independent variables without using derivatives;
2. *Calculus or gradient methods*, which use first- and sometimes second-order derivatives;
3. *Stochastic or evolutionary methods*, including genetic algorithms, simulated annealing, particle-swarm optimization, and neural-network-based methods.

3.4 Dynamic Optimization

Dynamic optimization explicitly considers the variation of the energy system with time. The system is generally represented by differential and algebraic equations.

The differential equations describe the evolution of the system states, including mass and energy balances, while the algebraic equations describe thermodynamic relations and other physical constraints.

A general dynamic optimization problem is written as

$$\underset{\mathbf{z}(t),\mathbf{y}(t),\mathbf{u}(t),t_f,\mathbf{w}}{\text{minimize}} \quad J\left[\mathbf{z}(t_f), \mathbf{y}(t_f), \mathbf{u}(t_f), t_f, \mathbf{w}\right], \tag{7}$$

subject to the differential–algebraic equality constraints

$$\mathbf{H}\left(\dot{\mathbf{z}}(t), \mathbf{z}(t), \mathbf{y}(t), \mathbf{u}(t), t, \mathbf{w}\right) = \mathbf{0}, \tag{8}$$

and the differential–algebraic inequality constraints

$$\mathbf{G}\left(\dot{\mathbf{z}}(t), \mathbf{z}(t), \mathbf{y}(t), \mathbf{u}(t), t, \mathbf{w}\right) \leq \mathbf{0}. \tag{9}$$

The initial state is specified by

$$\mathbf{z}(0) = \mathbf{z}^0. \tag{10}$$

Additional point conditions may be imposed at selected times t_s:

$$\mathbf{P}_s\left(\mathbf{z}(t_s), \mathbf{y}(t_s), \mathbf{u}(t_s), t_s, \mathbf{w}\right) = \mathbf{0}, \qquad t_s \in \left[t_0, t_f\right]. \tag{11}$$

The variables are also restricted by lower and upper bounds:

$$\mathbf{z}^L \leq \mathbf{z}(t) \leq \mathbf{z}^U, \tag{12a}$$

$$\mathbf{y}^L \leq \mathbf{y}(t) \leq \mathbf{y}^U, \tag{12b}$$

$$\mathbf{u}^L \leq \mathbf{u}(t) \leq \mathbf{u}^U, \tag{12c}$$

$$\mathbf{w}^L \leq \mathbf{w} \leq \mathbf{w}^U, \tag{12d}$$

$$\underline{t}_f \leq t_f \leq \overline{t}_f. \tag{12e}$$

Here,

- J is the scalar objective functional;
- $\mathbf{z}(t)$ is the differential-state profile vector;
- $\mathbf{y}(t)$ is the algebraic-state profile vector;
- $\mathbf{u}(t)$ is the control-variable profile vector;
- $\mathbf{w}$ is the vector of time-independent variables;
- t_f is the final time.

A common continuous-time objective functional is the Bolza form:

$$J\left[\mathbf{z}(t_f), \mathbf{y}(t_f), \mathbf{u}(t_f), t_f, \mathbf{w}\right] = Q\left(\mathbf{z}(t_f), \mathbf{y}(t_f), t_f, \mathbf{w}\right) + \int_{t_0}^{t_f} F\left(\mathbf{z}(t), \mathbf{y}(t), \mathbf{u}(t), t, \mathbf{w}\right) dt. \tag{13}$$

The function Q represents the contribution evaluated at the end of the optimization interval, whereas F represents contributions accumulated throughout the time horizon.

When the problem is discretized into N time intervals, the continuous integral is replaced by a summation:

$$\underset{\mathbf{z},\mathbf{y},\mathbf{u},t_f,\mathbf{w}}{\text{minimize}} \quad J\left[\mathbf{z}, \mathbf{y}, \mathbf{u}, t_f, \mathbf{w}\right] = Q\left(\mathbf{z}_N, \mathbf{y}_N, N, \mathbf{w}\right) + \sum_{n=1}^{N} F\left(\mathbf{z}_n, \mathbf{y}_n, \mathbf{u}_n, n, \mathbf{w}\right). \tag{14}$$

Dynamic optimization methods are divided into two main groups:

1. *Indirect methods*, including calculus of variations and dynamic programming;
2. *Direct methods*, in which the continuous-time problem is discretized and solved numerically. Direct methods can be sequential or simultaneous.

3.5 Intertemporal Static Optimization

Intertemporal static optimization provides an intermediate approach between single-point static optimization and fully dynamic optimization.

The actual operating profile over a selected period, such as a day, week, year, or complete life cycle, is divided into a number of time intervals. Each interval is approximated as a steady-state operating condition, and transient periods between the intervals are neglected.

The static optimization problem represented by Eqs. (1)–(5) is solved separately for each time interval. The optimized intervals are then combined within an upper-level synthesis or design optimization problem.

This method assumes that operation during one time interval does not affect operation during the other intervals. When this assumption is invalid, a fully dynamic optimization formulation is required.

In the proposed research, the intertemporal static optimization framework is applied to the operation of a grid-connected solar photovoltaic (PV)–battery–load microgrid by dividing the operating horizon into discrete time intervals, such as hourly periods over a 24-hour day. Within each interval, the microgrid is assumed to operate under steady-state conditions, where the photovoltaic power generation, load demand, electricity purchased from the utility grid, electricity exported to the grid, and battery charging or discharging rates are treated as constant. The optimization simultaneously determines the optimal power flow for all time intervals while satisfying the power balance and operational constraints of the microgrid. Unlike conventional static optimization, however, the battery State of Charge (SOC) links consecutive time intervals, as the available stored energy at each interval depends on the charging and discharging decisions made during the previous interval. Consequently, although each interval is represented as a steady-state operating condition, the overall optimization remains intertemporally coupled through the battery dynamics. This formulation enables the proposed Energy Management System (EMS) to determine an optimal battery scheduling strategy that minimizes electricity imported from the utility grid, maximizes renewable electricity exported to the grid, maximizes renewable energy utilization, maximizes microgrid profit, and minimizes operating cost over the entire optimization horizon.

The optimization problem is formulated over a sequence of steady-state time intervals. While the operating conditions within each interval are assumed to be static, the intervals are interconnected through the battery State of Charge (SOC), which evolves according to the charging and discharging decisions. This allows the optimization to capture the intertemporal characteristics of battery energy storage without requiring a fully dynamic system model.

4 Steps in Formulating An Optimization Problem[12]

4.1 System Boundaries

The first step in any optimization problem is to define the **system boundaries**. All subsystems that affect system performance should be included. When the system is overly complex, it is often desirable to divide it into smaller subsystems. In this case, it is often reasonable to carry out optimization on each subsystem independently; that is, *sub-optimization* of the subsystems is performed.

4.2 Objective Functions and System Criteria

The next step in an optimization problem is to define the **system criteria**, which are sometimes called **objective functions**. An objective function is based on the desire or purpose of the decision maker, and it can be either maximized or minimized. Optimization criteria can vary widely. For instance, optimization criteria can be based on:

- **Economic aims:** total capital investment, total annual levelized costs, cost of exergy destruction, and cost of environmental impact.
- **Efficiency aims:** energy efficiency, exergy efficiency, and other efficiency measures.
- **Technological goals:** production rate, production time, and total weight.
- **Environmental objectives:** reduction of pollutant emissions.
- **Other objectives:** any additional performance measures relevant to the system.

Note that more than one objective function can be considered when solving an optimization problem through **multi-objective optimization**.

4.3 Decision Variables

Another essential step in formulating an optimization problem is the selection of independent **decision variables**

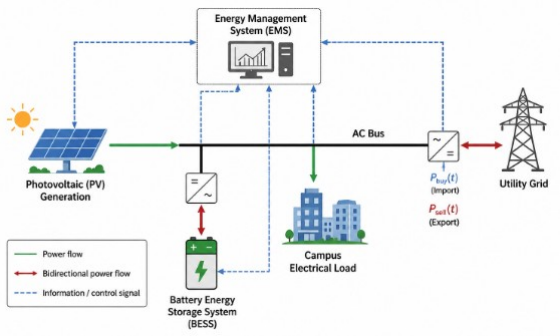


Figure 1: *Configuration of the grid-connected solar PV-Battery-Microgrid*

that adequately characterize the possible design options. To select decision variables, it is important to:

(a) Include all important variables that can affect the performance and cost-effectiveness of the system.
(b) Exclude variables of minor importance.
(c) Distinguish among independent variables whose values are amenable to change.

In a given optimization problem, only the decision variables are changed. Variables whose values are calculated from the independent variables using mathematical models are referred to as **dependent variables**.

4.4 Constraints

The **constraints** in a design problem arise due to limitations on the ranges of the physical variables, basic conservation principles that must be satisfied, and other physical or operational restrictions imposed on the system.

5 Problem FOrmulation

5.1 Microgrid COnfiguration

This research considers a grid-connected solar photovoltaic (PV)–battery–load microgrid, as illustrated in Fig. 1. The microgrid consists of a photovoltaic generation system, a Battery Energy Storage System (BESS), local electrical loads, and a bidirectional connection to the utility grid. The Energy Management System (EMS) determines the optimal power exchange among these components over a predefined optimization horizon.

During each time interval, electricity generated by the PV system is first supplied to meet the local load demand. Surplus photovoltaic energy is then stored in the battery whenever storage capacity is available. When the battery reaches its operational limit and excess renewable energy remains, the surplus electricity is exported to the utility grid to maximize renewable energy utilization and generate additional revenue. Conversely, when photovoltaic generation is insufficient to satisfy the load demand, the EMS determines the optimal battery discharge strategy before purchasing electricity from the utility grid. This operational strategy prioritizes renewable energy utilization while reducing dependence on grid electricity and improving the economic performance of the microgrid.

The proposed EMS explicitly considers the utility grid as an active participant in the optimization. Consequently, the battery is scheduled not only to satisfy local demand but also to support strategic electricity trading through optimal charging, discharging, and electricity export decisions.

5.2 Intertemporal Static Optimization Formulation

The proposed optimization problem is formulated as an intertemporal static optimization problem over a finite time horizon. The operating period, such as a 24-hour day, is divided into discrete hourly intervals, with each interval approximated as a steady-state operating condition. Within each interval, photovoltaic generation, load demand, battery charging and discharging power, and electricity exchanged with the utility grid are assumed to remain constant.

Although each interval is represented by a steady-state operating condition, the intervals are interconnected through the battery State of Charge (SOC). The available battery energy at the beginning of each interval depends on the charging and discharging decisions made during the previous interval, thereby introducing intertemporal coupling without requiring a fully dynamic system model. This formulation captures the long-term operational behaviour of the battery while maintaining a computationally efficient optimization framework.

The optimization horizon is defined as

t=1,2,. . . ,T,

where T represents the total number of time intervals within the optimization period.

5.3 Decision Variables

A possible set of decision variables for each time t in the optimization problem for the optimization to determine is in Table 1. The decision variables collectively define the optimal power flow within the microgrid while satisfying all operational constraints.

Table 1: *Decision Variables of the Proposed Optimization Model*

Variable	Description
$P_{\text{buy}}(t)$	Electricity imported from the utility grid (kW)
$P_{\text{sell}}(t)$	Electricity exported to the utility grid (kW)
$P_{\text{ch}}(t)$	Battery charging power (kW)
$P_{\text{dis}}(t)$	Battery discharging power (kW)
$SOC(t)$	Battery state of charge (%)
$P_{\text{PV}}(t)$	Photovoltaic power generation (kW)
$P_{\text{Load}}(t)$	Microgrid load demand (kW)

6 Objective Functions

The proposed Energy Management System is formulated as a multi-objective optimization problem. Unlike conventional EMS strategies that focus primarily on operating cost reduction, the proposed framework simultaneously considers both the economic interests of the microgrid owner and support for the utility grid. The optimization objectives are the following.

$$\min \sum_{t=1}^{T} P_{\text{buy}}(t) \tag{1}$$

to minimize electricity imported from the utility grid,

$$\max \sum_{t=1}^{T} P_{\text{sell}}(t) \tag{2}$$

to maximize renewable electricity exported to the utility grid,

$$\max \sum_{t=1}^{T} P_{\text{PV,used}}(t) \tag{3}$$

to maximize renewable energy utilization,

$$\max \left(\sum_{t=1}^{T} \left[\lambda_{\text{sell}}(t) P_{\text{sell}}(t) - \lambda_{\text{buy}}(t) P_{\text{buy}}(t) \right] \right) \tag{4}$$

to maximize microgrid profit, and

$$\min C_{\text{op}} \tag{5}$$

to minimize the total operating cost of the microgrid.

These objectives are solved simultaneously subject to the operational constraints presented in the following section.

7 Mathematical Formulation of the Optimization Problem

7.1 Objective Functions

The proposed Energy Management System (EMS) is formulated as a multi-objective optimization problem that simultaneously considers the economic performance of the microgrid owner and support for the utility grid. Unlike conventional EMS strategies that primarily minimize electricity costs, the proposed framework explicitly incorporates electricity trading and renewable energy utilization as primary optimization objectives.

Objective 1: Minimize Electricity Imported from the Utility Grid Purchasing electricity from the utility grid increases the operating expenditure of the microgrid while generating revenue for the utility provider. Therefore, the first objective is to minimize the total electricity imported over the optimization horizon,

$$\min f_1 = \sum_{t=1}^{T} P_{\text{buy}}(t) \tag{6}$$

where $P_{\text{buy}}(t)$ denotes the electricity imported from the utility grid during time interval t, and T represents the total number of optimization intervals.

Reducing grid imports decreases the dependence of the microgrid on external electricity supply while improving its operational autonomy.

Objective 2: Maximize Renewable Electricity Exported to the Utility Grid When photovoltaic generation exceeds the local load demand and the battery has sufficient reserve capacity, surplus renewable electricity should be exported to the utility grid. Selling excess electricity generates additional revenue for the microgrid owner while supporting renewable energy penetration within the power system.

The second objective is therefore expressed as

$$\max f_2 = \sum_{t=1}^{T} P_{\text{sell}}(t) \tag{7}$$

where $P_{\text{sell}}(t)$ denotes the electricity exported to the utility grid.

Unlike many existing optimization studies, electricity export is treated as a primary optimization objective rather than merely a consequence of economic dispatch.

Objective 3: Maximize Renewable Energy Utilization To improve the sustainability of the microgrid, renewable electricity generated by the photovoltaic system should be utilized whenever possible rather than curtailed. This objective is expressed as

$$\max f_3 = \sum_{t=1}^{T} P_{\text{PV,use}}(t) \tag{8}$$

or equivalently,

$$\min \sum_{t=1}^{T} P_{\text{curtail}}(t). \tag{9}$$

where $P_{\text{PV,use}}(t)$ is the photovoltaic power utilized by the microgrid and $P_{\text{curtail}}(t)$ is the curtailed renewable energy.

Objective 4: Maximize Microgrid Profit The economic performance of the microgrid is evaluated by considering both the revenue obtained from selling electricity to the utility grid and the cost incurred from purchasing electricity.

The total profit is given by

$$Profit = Revenue - Cost \tag{10}$$

where

$$Revenue = \sum_{t=1}^{T} \lambda_{\text{sell}}(t) P_{\text{sell}}(t) \tag{11}$$

and

$$Cost = \sum_{t=1}^{T} \lambda_{\text{buy}}(t) P_{\text{buy}}(t). \tag{12}$$

Therefore, the profit maximization objective is

$$\max f_4 = \sum_{t=1}^{T} \left[\lambda_{\text{sell}}(t) P_{\text{sell}}(t) - \lambda_{\text{buy}}(t) P_{\text{buy}}(t) \right]. \tag{13}$$

where $\lambda_{\text{buy}}(t)$ and $\lambda_{\text{sell}}(t)$ denote the electricity buying and selling prices, respectively.

Objective 5: Minimize Operating Cost The final objective minimizes the overall operating cost of the microgrid, including electricity purchase costs and battery degradation costs.

$$\min f_5 = C_{\text{grid}} + C_{\text{battery}} \tag{14}$$

where

$$C_{\text{grid}} = \sum_{t=1}^{T} \lambda_{\text{buy}}(t) P_{\text{buy}}(t) \tag{15}$$

and

$$C_{\text{battery}} = \sum_{t=1}^{T} C_{\text{deg}}(t). \tag{16}$$

Here, $C_{\text{deg}}(t)$ represents the battery degradation or operational cost associated with battery cycling.

Multi-objective Optimization Problem The complete optimization problem is formulated as

$$\begin{aligned} &\text{Minimize} \quad f_1,\ f_5 \\ &\text{Maximize} \quad f_2,\ f_3,\ f_4 \end{aligned} \tag{17}$$

subject to the operational constraints presented in the following section.

8 Optimization Constraints and Solution Methodology

8.1 Power Balance Constraint

The Energy Management System must satisfy the power balance during every optimization interval. According to the principle of energy conservation, the electrical power supplied by the photovoltaic system, battery, and utility grid must equal the electrical power consumed by the load, battery charging process, and electricity exported to the utility grid.

$$P_{\text{PV}}(t) + P_{\text{buy}}(t) + P_{\text{dis}}(t) = P_{\text{Load}}(t) + P_{\text{ch}}(t) + P_{\text{sell}}(t), \quad \forall t \tag{18}$$

where

- $P_{\text{PV}}(t)$: photovoltaic power generation,
- $P_{\text{Load}}(t)$: load demand,
- $P_{\text{buy}}(t)$: electricity imported from the grid,
- $P_{\text{sell}}(t)$: electricity exported to the grid,
- $P_{\text{ch}}(t)$: battery charging power,
- $P_{\text{dis}}(t)$: battery discharging power.

8.2 Battery State-of-Charge Constraint

The battery State of Charge (SOC) evolves according to the charging and discharging processes,

$$SOC(t+1) = SOC(t) + \eta_c P_{\text{ch}}(t) \Delta t - \frac{P_{\text{dis}}(t) \Delta t}{\eta_d}. \tag{19}$$

where η_c and η_d denote the charging and discharging efficiencies, respectively.

8.3 Battery Capacity Constraint

The battery State of Charge must satisfy

$$SOC_{\min} \le SOC(t) \le SOC_{\max}, \quad \forall t. \tag{20}$$

This prevents battery overcharging and deep discharge.

8.4 Battery Charging and Discharging Constraints

The charging and discharging power are constrained by

$$0 \le P_{\text{ch}}(t) \le P_{\text{ch}}^{\max} \tag{21}$$

and

$$0 \le P_{\text{dis}}(t) \le P_{\text{dis}}^{\max}. \tag{22}$$

To prevent simultaneous charging and discharging,

$$P_{\text{ch}}(t) P_{\text{dis}}(t) = 0, \qquad \forall t. \tag{23}$$

8.5 Grid Import and Export Constraints

The power exchanged with the utility grid is constrained by

$$0 \le P_{\text{buy}}(t) \le P_{\text{buy}}^{\max} \tag{24}$$

and

$$0 \le P_{\text{sell}}(t) \le P_{\text{sell}}^{\max}. \tag{25}$$

These limits represent the capacity of the grid connection.

8.6 Peak-Shifting Strategy

Peak shifting is incorporated into the proposed Energy Management System to improve the economic performance of electricity trading while supporting the utility grid. During periods of high photovoltaic generation, surplus renewable energy is preferentially stored in the battery. Once the battery reaches its desired operating reserve, the remaining surplus electricity is exported to the utility grid. Conversely, during periods of high electricity prices or increased load demand, the battery is discharged to reduce electricity purchased from the utility grid.

This strategy reduces peak demand, increases renewable energy utilization, and improves the profitability of the microgrid while supporting more efficient operation of the utility grid.

8.7 Solution Methodology

The proposed optimization problem is solved over a 24-hour optimization horizon using an intertemporal static optimization framework. Initially, representative photovoltaic generation and electrical load profiles are used to validate the optimization model. Subsequently, actual photovoltaic generation and electricity demand data obtained from TEMS the University of Tsukuba Electricity Monitoring System (TEMS)the University of Tsukuba campus will be incorporated to evaluate the proposed Energy Management System under realistic operating conditions.

The optimization results will be assessed using several performance indicators, including total electricity imported from the utility grid, renewable electricity exported to the utility grid, renewable energy utilization, operating cost, microgrid profit, and battery State of Charge profile. Finally, the proposed framework is designed to accommodate future integration of Vehicle-to-Grid (V2G) technology by incorporating electric vehicles as additional distributed energy storage resources.

9 Optimization Workflow

Following the formulation of the multi-objective optimization problem, the proposed Energy Management System (EMS) will be implemented to determine the optimal operating strategy of a grid-connected solar photovoltaic (PV)–battery–load microgrid. The optimization process consists of data acquisition, mathematical formulation, optimization, and performance evaluation.

Initially, representative photovoltaic generation, electrical load demand, and electricity price data will be collected over the selected optimization horizon. During the preliminary stage of this research, publicly available datasets and representative solar irradiation profiles will be employed to verify the mathematical formulation and validate the optimization model. Subsequently, the developed framework will be applied to actual electricity demand and photovoltaic generation data obtained from the University of Tsukuba campus, allowing the proposed Energy Management System to be evaluated under realistic operating conditions.

The collected data are first pre-processed to remove missing values and synchronize the sampling interval before being incorporated into the optimization model. The decision variables, objective functions, and operational constraints presented in Sections V and VI are then formulated within an intertemporal static optimization framework. The optimization algorithm determines the optimal battery charging and discharging schedule, electricity imported from the utility grid, and renewable electricity exported to the utility grid for every optimization interval while satisfying all technical and operational constraints.

The resulting optimal operating schedule is evaluated using several performance indicators, including electricity imported from the utility grid, renewable electricity exported to the utility grid, renewable energy utilization, operating cost, and microgrid profit. These performance metrics are compared with a conventional battery scheduling strategy to quantify the economic and operational improvements achieved by the proposed optimization framework. Furthermore, we will study in detail about self-suffiency (percentage indicating minimising import to the grid) and self-consumption of the microgrid and later evaluate these ratios to measure battery and microgrid performance.

10 Conclusion

This research proposes a multi-objective optimization framework for the energy management of a grid-connected solar photovoltaic (PV)–battery–load microgrid. Motivated by the increasing integration of renewable energy and the need for more intelligent battery scheduling, the proposed framework seeks to balance the economic interests of the microgrid owner with support for the utility grid. Unlike conventional energy management strategies that primarily minimize operating costs or maximize self-consumption, the proposed optimization explicitly incorporates electricity trading by simultaneously minimizing electricity imported from the utility grid, maximizing renewable electricity exported to the grid, maximizing renewable energy utilization, maximizing microgrid profit, and minimizing operating costs.

The optimization problem is formulated as an intertemporal static optimization model in which the operating horizon is divided into discrete time intervals while the battery State of Charge (SOC) provides the temporal linkage between successive intervals. This formulation establishes the mathematical foundation for determining the optimal battery charging and discharging schedule while satisfying the operational constraints of the microgrid.

The proposed optimization model will first be validated using representative photovoltaic generation and electrical load profiles to verify the mathematical formulation. Subsequently, the framework will be implemented using actual electricity demand and photovoltaic generation data from the University of Tsukuba campus to evaluate its effectiveness under realistic operating conditions. Future work will also investigate the integration of Vehicle-to-Grid (V2G) technology, enabling electric vehicles to participate as distributed energy storage resources within the same optimization framework.

It is expected that the proposed Energy Management System will provide a practical and scalable approach for improving renewable energy utilization, reducing dependence on the utility grid, increasing microgrid profitability, and supporting the development of sustainable campus microgrids.

11 Usage of Generative AI

The optimisation problem, objective functions, some of the constraints, the optimisation workflow, and the plan to use campus data has been decided from discussing and learning from professor and lab senpais/seniors and reading papers without AI. [7] has been obtained from suggestion by lab senpai. Generative AI has been used to help formulate the objective functions and constraints into mathematical formulas and constraints, get several good source papers and edit phrases.

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